

BAAZ JKBOSE



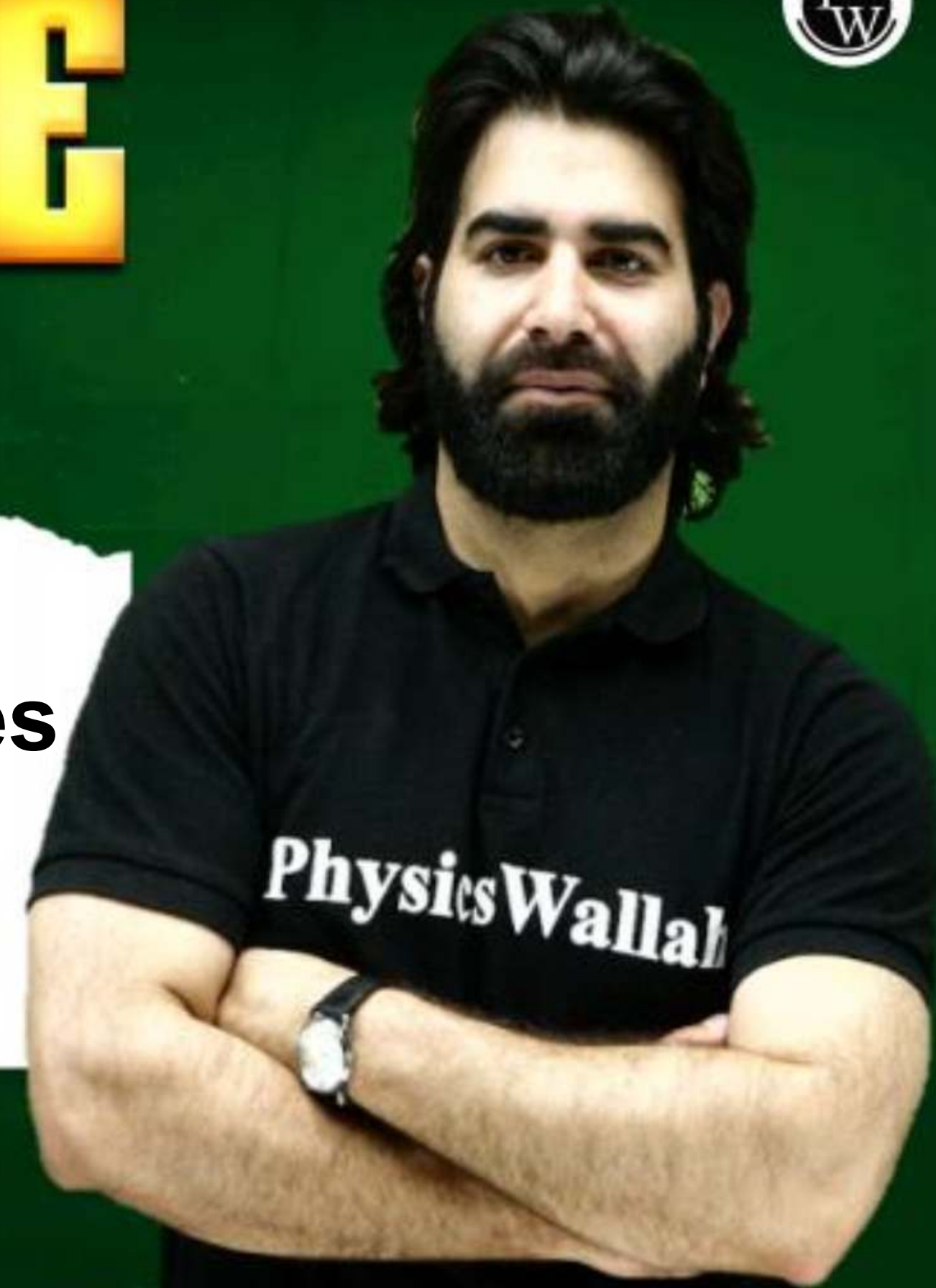
2026 CLASS-11TH

Breathing and Exchange of Gases

LECTURE-3

Zoology

Dr Adil Altaf sir



Topics To Be Covered

- 1 Mechanism of breathing
- 2 Lung volumes and capacity

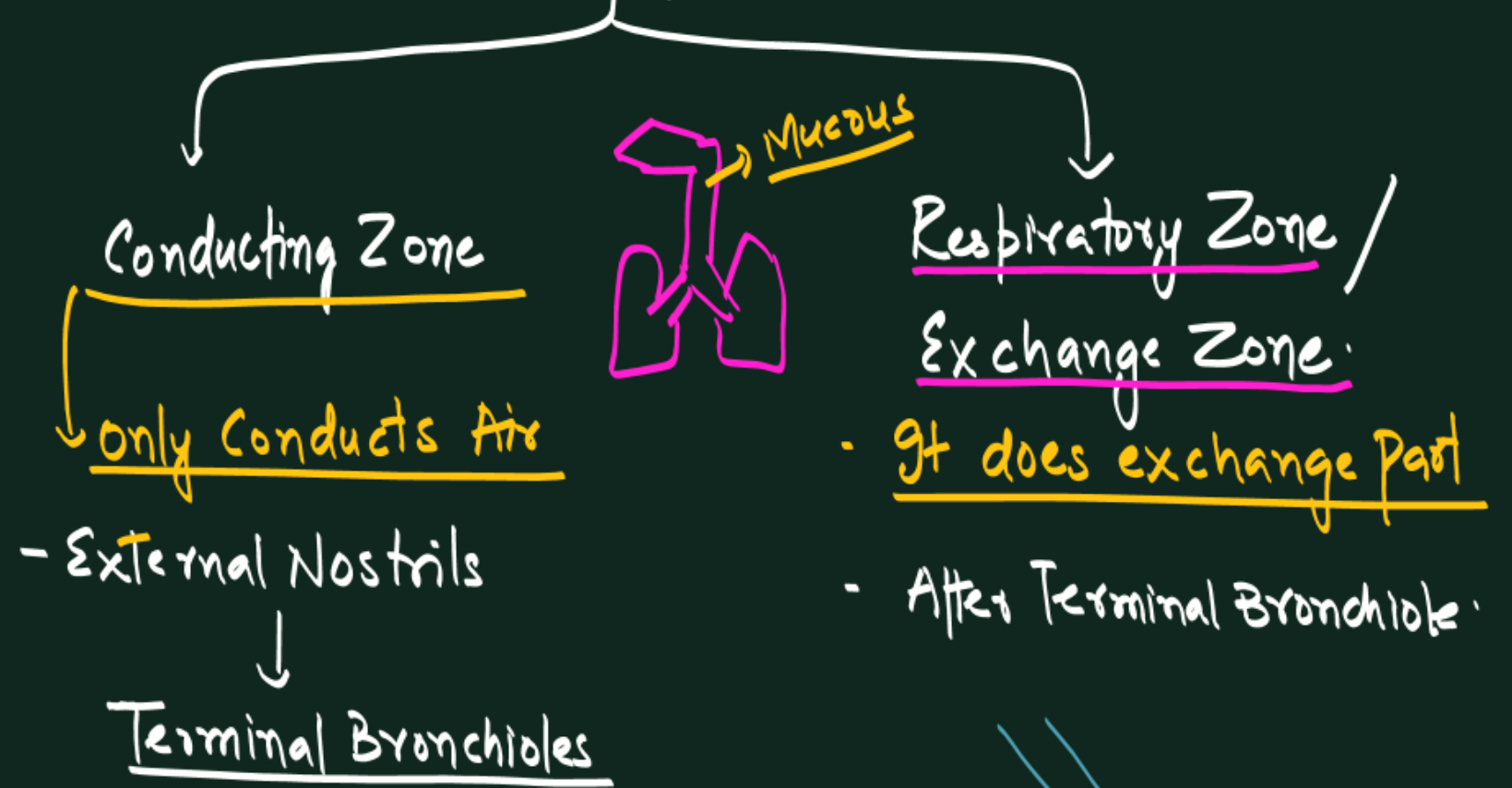
Revision

→ Human Respiratory System

- Lungs — covering pleura
- Alveoli — Main Epithelium Simple Sq.
- Trachea — Rings (complete / Incomplete)

↑ Evaporation
Surface

Respiratory System



→ Other Functions

- Air Purification — Removes Foreign Body
- Humidifies Air
- Warms air to Body Temp

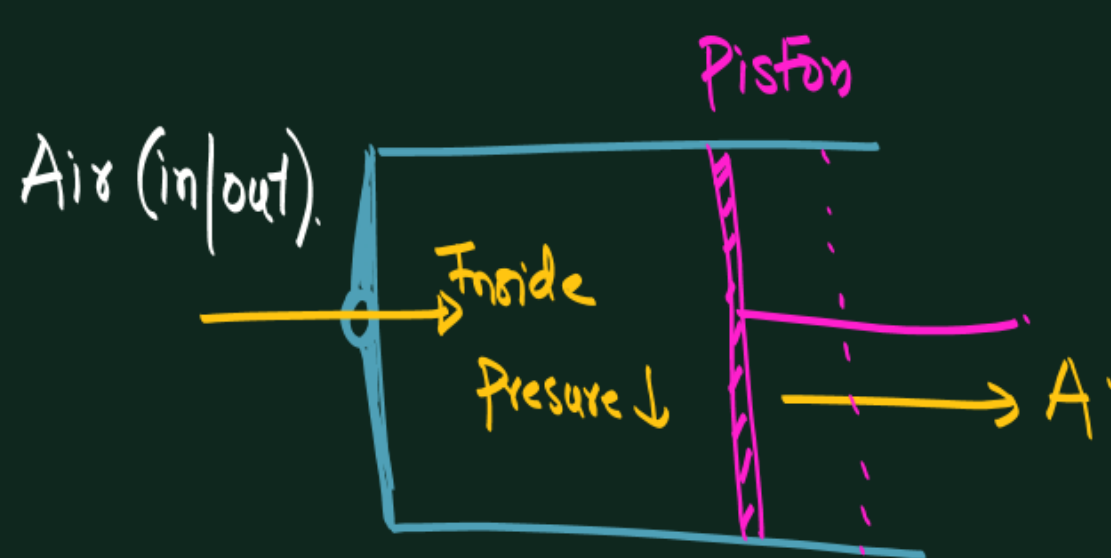
Mechanism of Breathing (Inspiration & Expiration)

→ Mechanism of Breathing is Based on difference in Intrapulmonary Pressure (pressure inside lungs) and Atmospheric (External) Pressure.

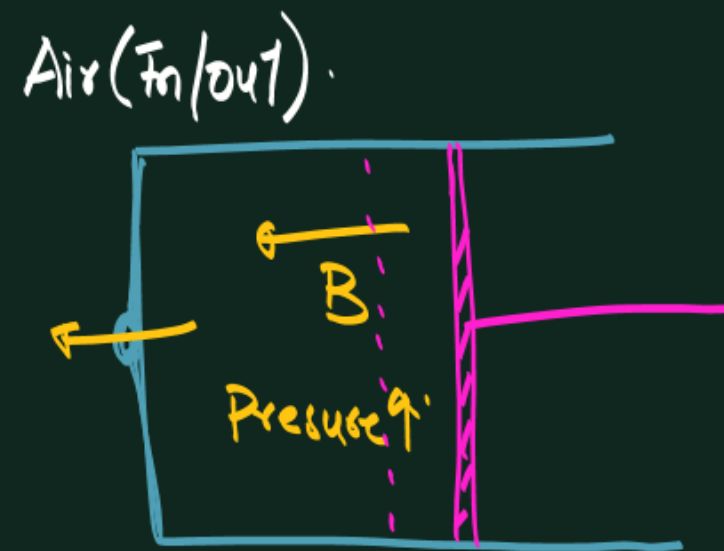
We know PV is Const $\Rightarrow$ Boyle's Law

$$PV = \text{Const}$$

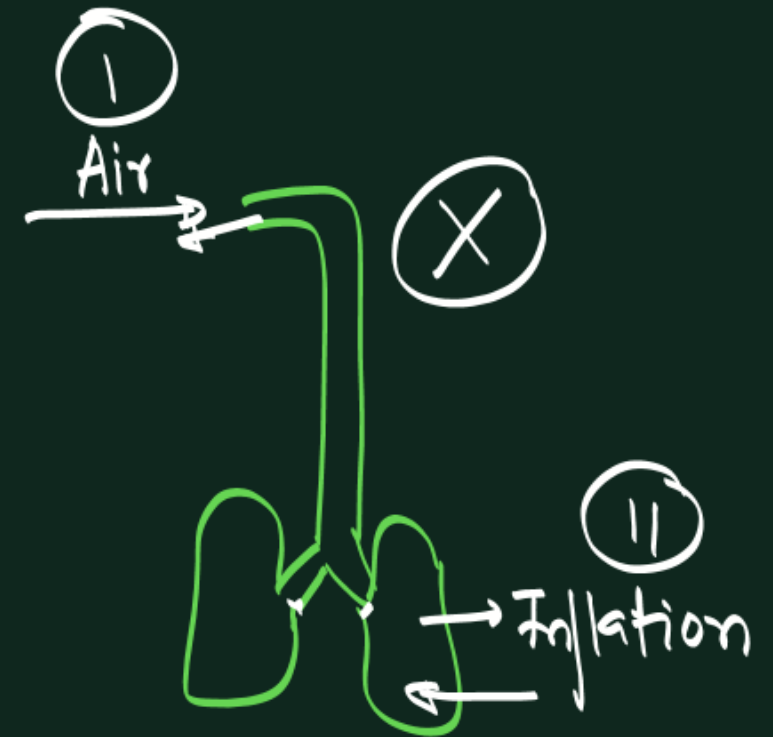
$V \uparrow, P \downarrow$ and vice versa.



Case I, Vol $\uparrow$, Pressure $\downarrow$



Case II, Vol $\downarrow$, Pressure $\uparrow$



For changing pressure, we need to change volume.

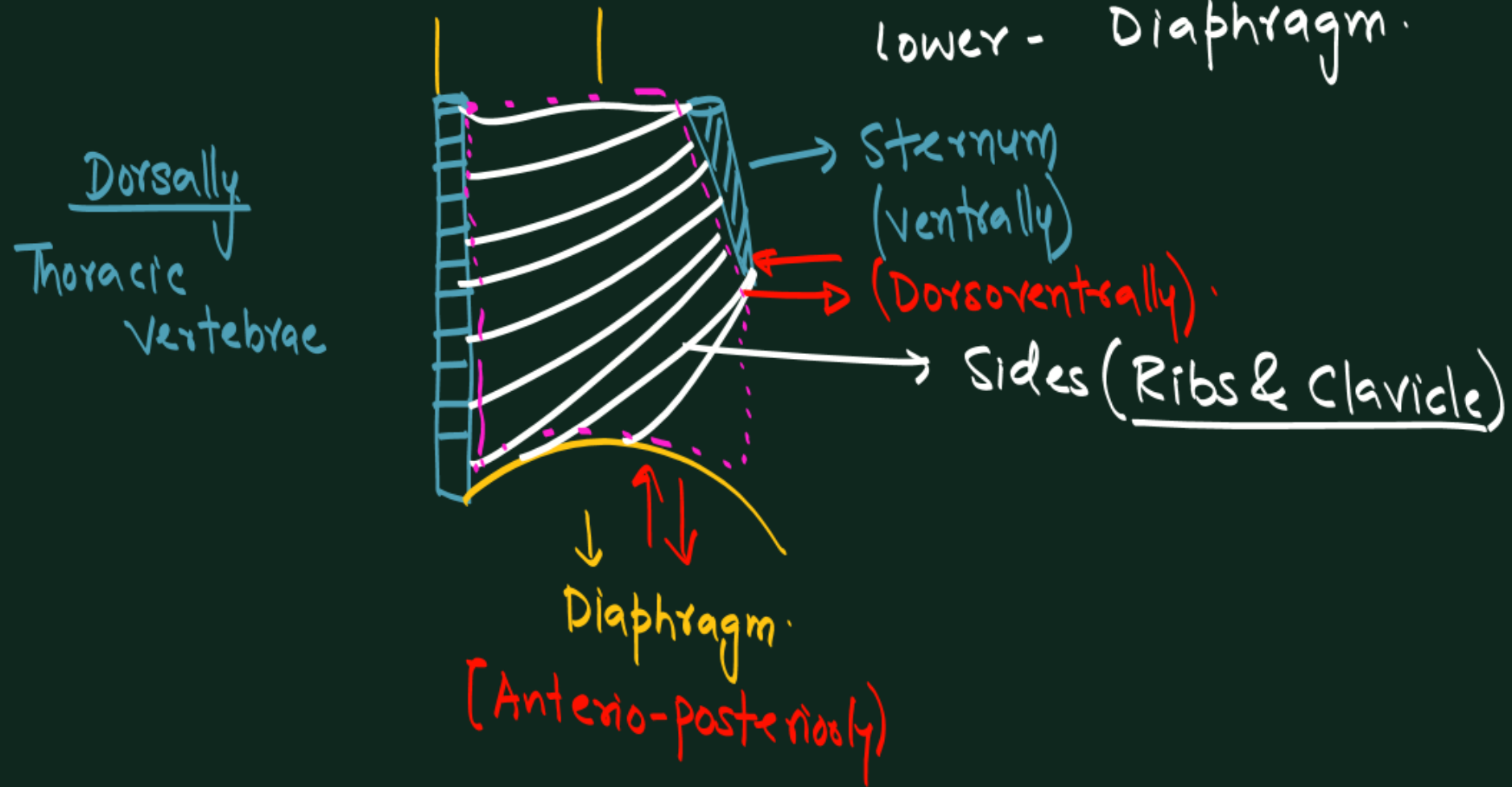
$V \downarrow P \uparrow$

$V \uparrow P \downarrow$

→ Change in Thoracic volume will bring change in Thoracic pressure.

→ Boundaries of Thoracic chamber

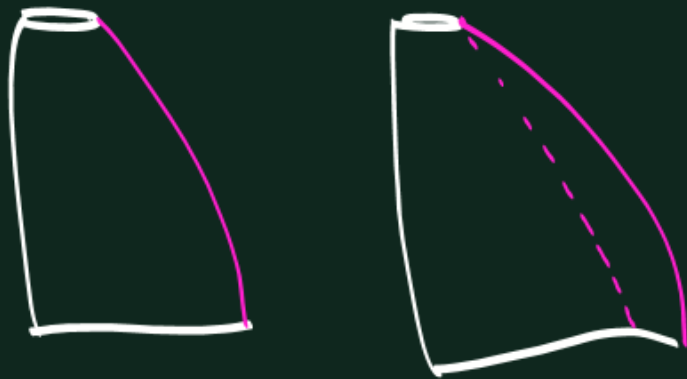
- Dorsal - V. Column (Thoracic vertebrae)
- Ventral - Sternum.
- Side - Ribs, Clavicle
- Lower - Diaphragm.



We can change Thoracic Volume by two ways.

① Lifting Ribs & Sternum.

- To Elevate Ribs and Sternum.
- Contraction of External Intercostal Muscles.
- Volume Increase (Dorsoventrally).

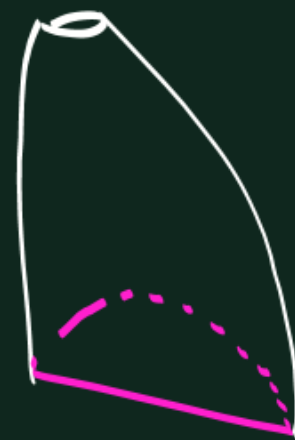


② By Contracting and Relaxing Diaphragm.

- Diaphragm when Relaxed - Dome Shaped.



- Diaphragm when contract, it become flat (Volume Increase)



Summary



① For Inspiration.

- Thoracic volume should Increase.
- This decreases intrapulmonary pressure.
- That means (Negative pressure in Lungs).
- For that, Ribs and sternum - Should Elevate.
- Because of Contraction of External ICM.
- Also Diaphragm should Contract,
Diaphragm becomes Flat.
- Hence, volume increases, pressure inside decreases and Air Moves from Atmosphere to the Lungs.

② For Expiration

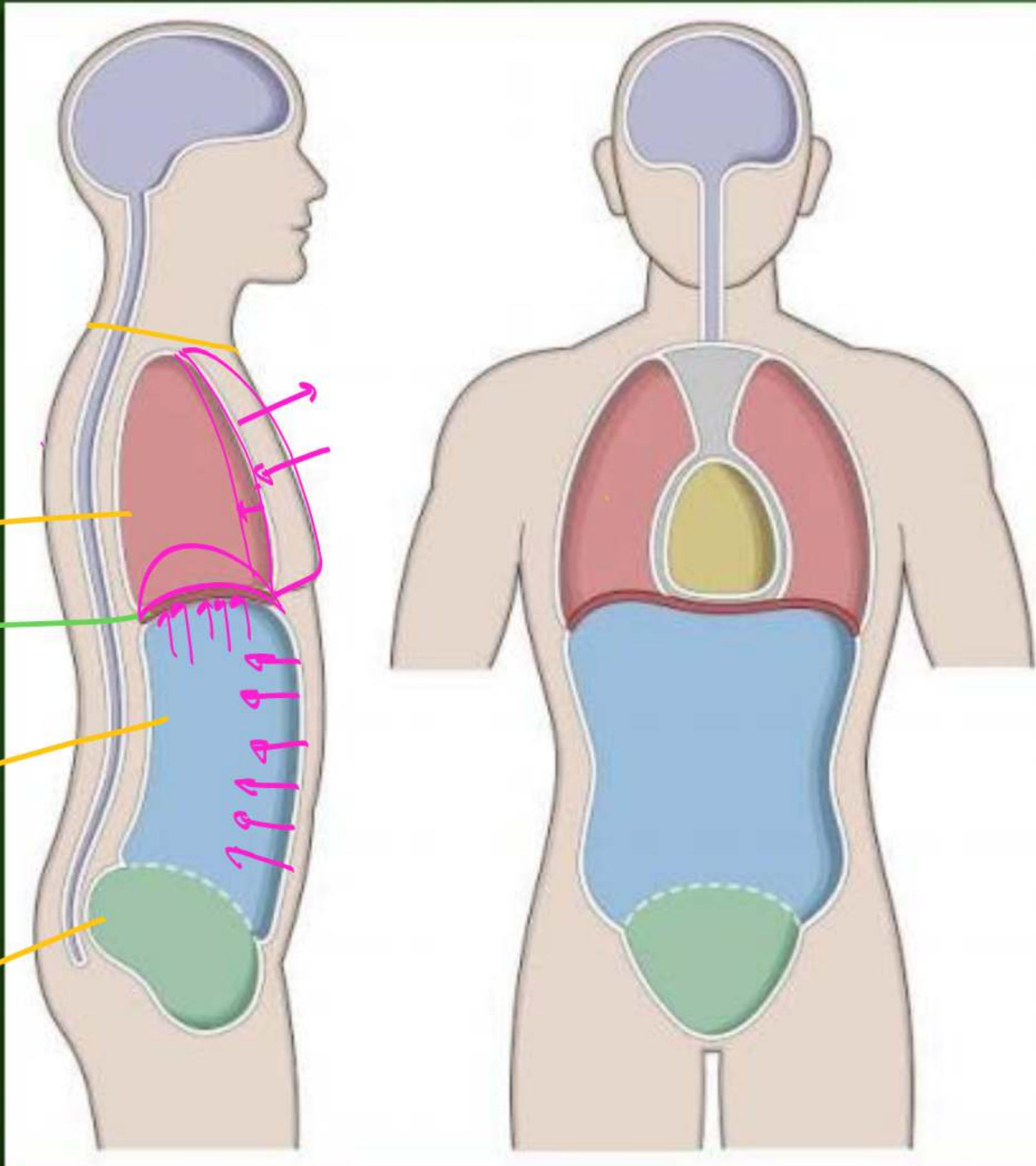
- Thoracic volume should decrease.
- This increases intrapulmonary pressure.
- That means positive pressure in Lungs.
- Ribs & Sternum should be back to Normal position, when when External ICM Relaxes.
- Also Diaphragm relaxes and becomes dome shaped.
- Hence volume decrease and pressure inside Lungs Increase, there Air will move from inside to outside.

Thoracic Cavity

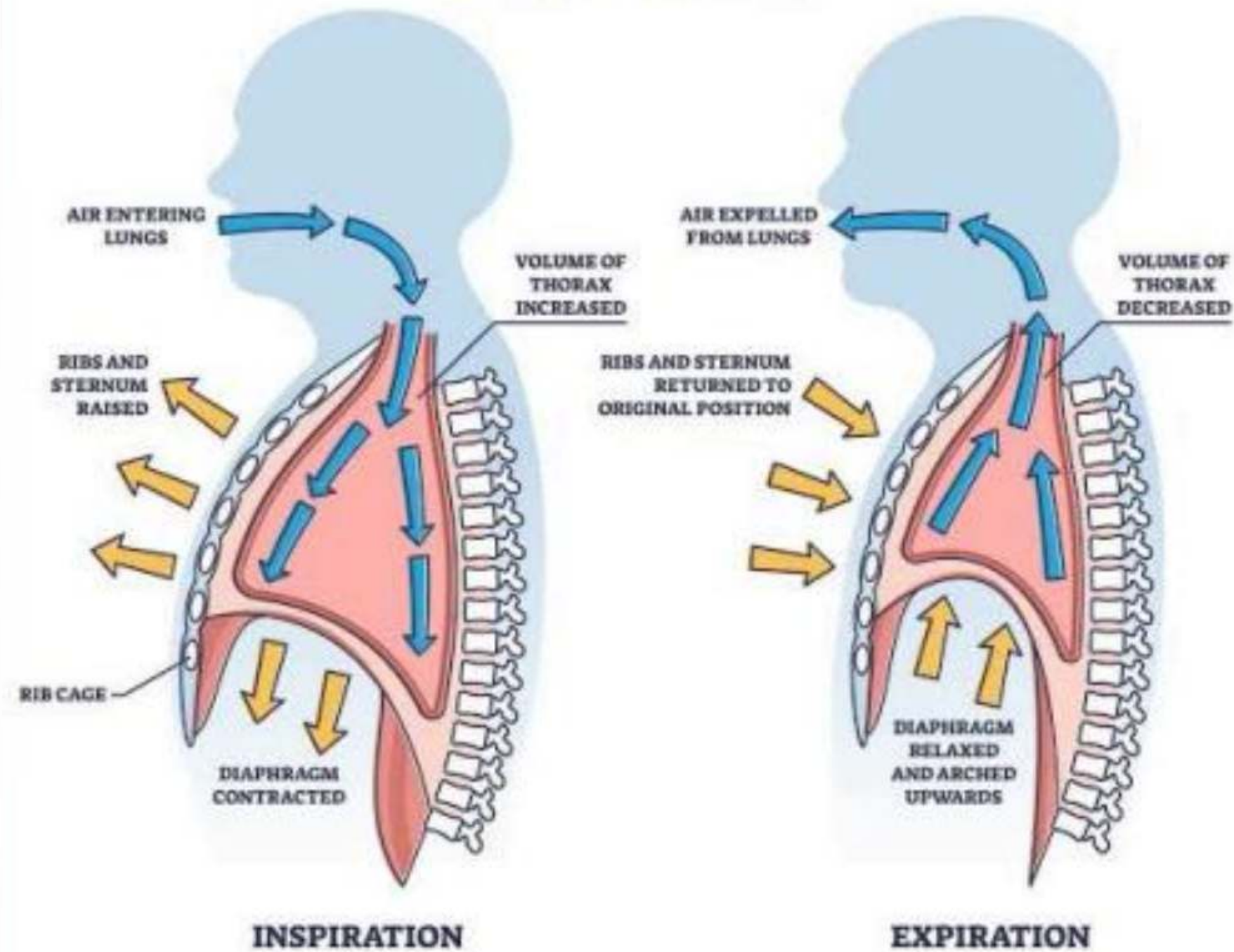
Diaphragm

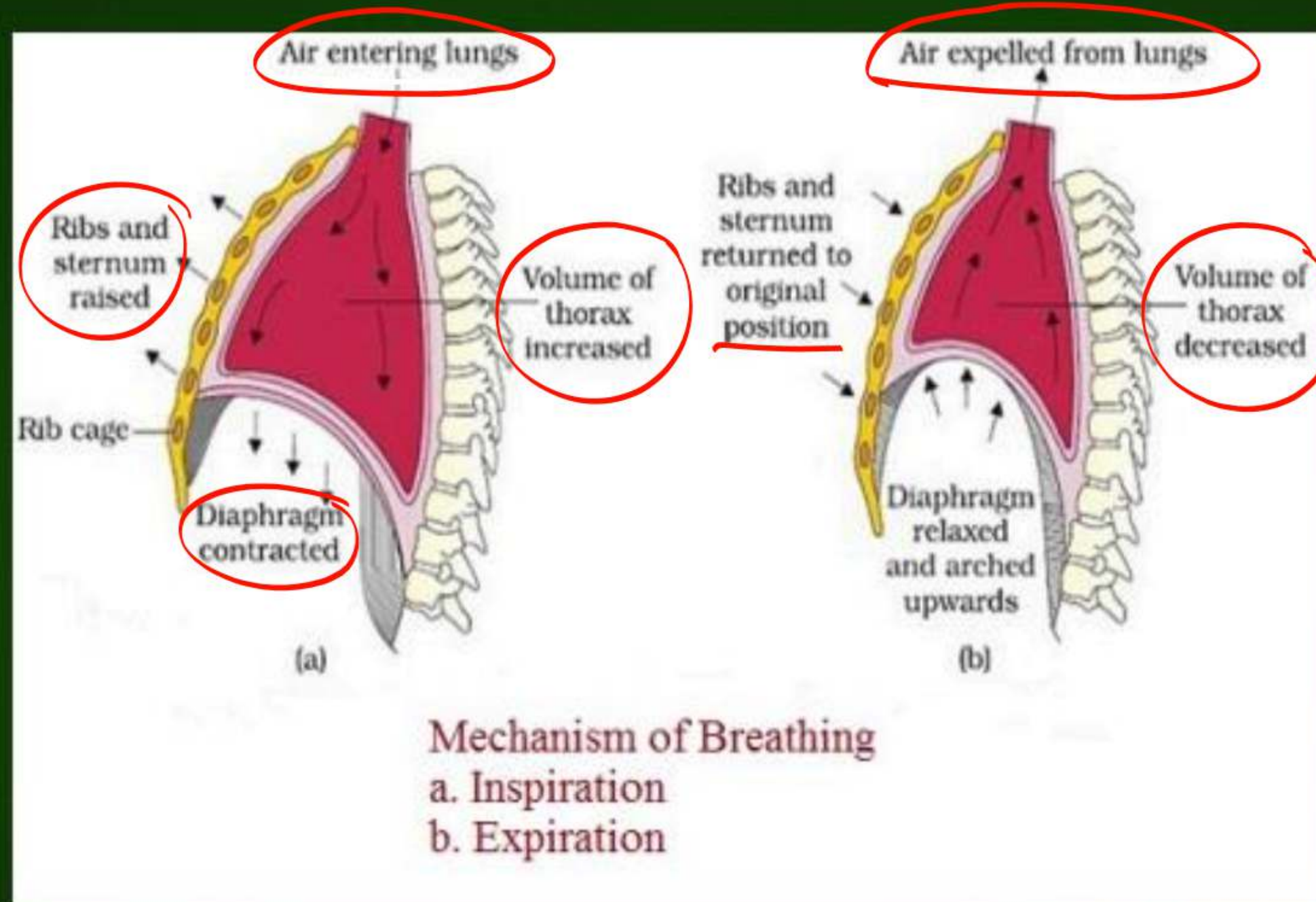
Abdominal Cavity

Pelvic Cavity



MECHANISM OF BREATHING



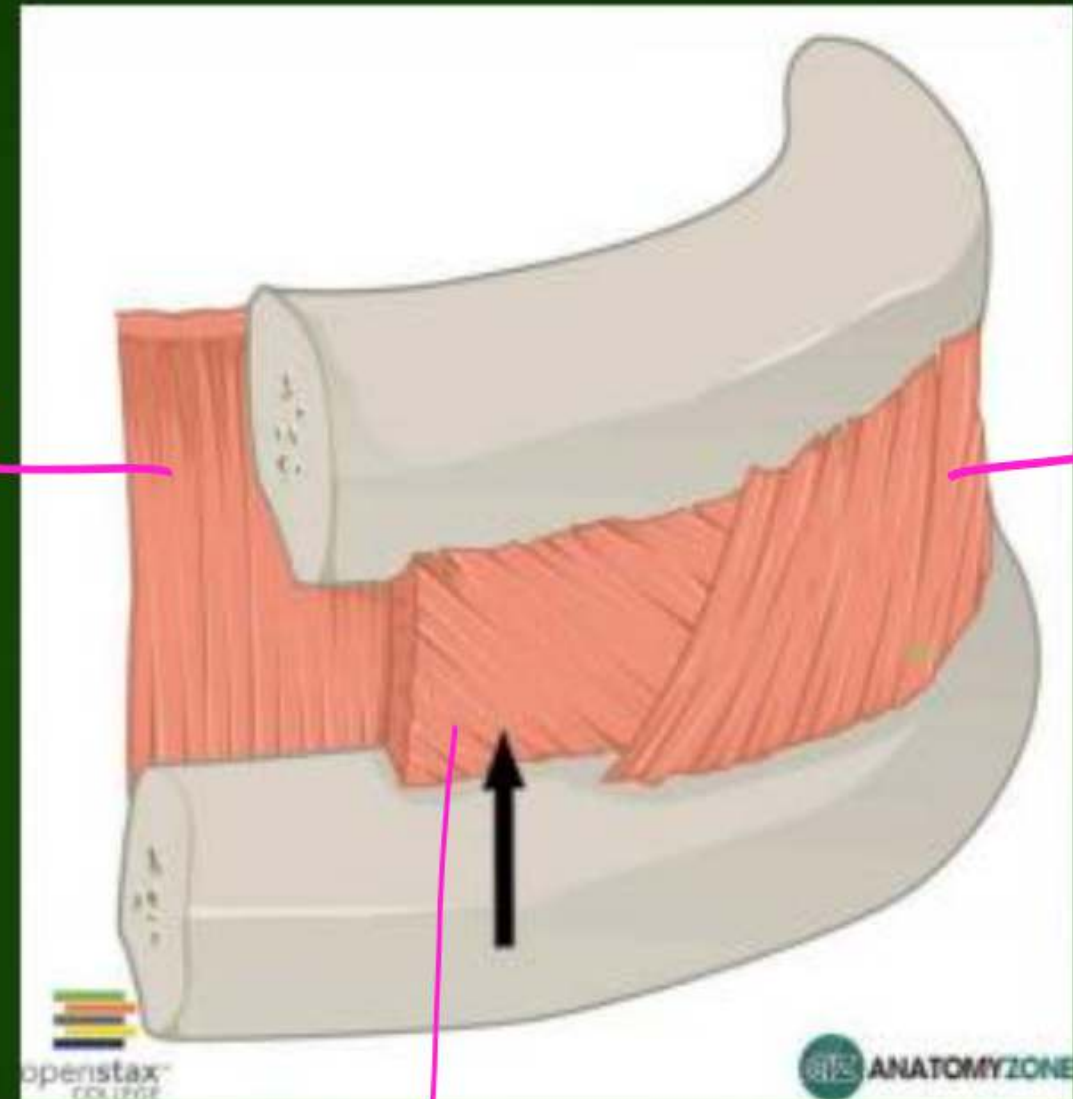
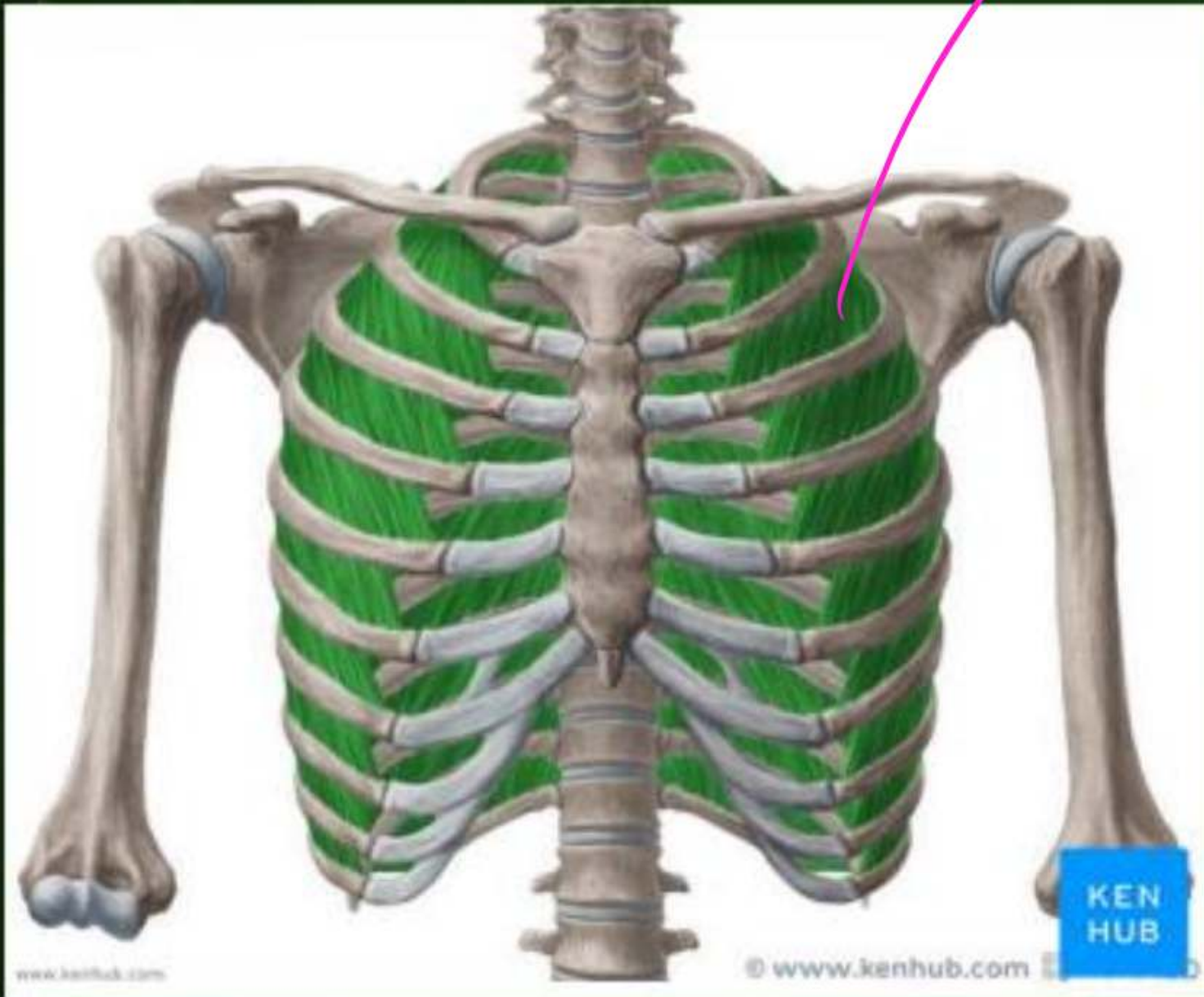


Home work

Role of Abdominal Muscle and

Internal Intercostal Muscles in Breathing

→ Forceful Expiration.



Thank
You